

Post-transition State Bifurcations in Reactions That Form Pimarenyl Cation Diastereomers—Implications for Terpene Biosynthesis

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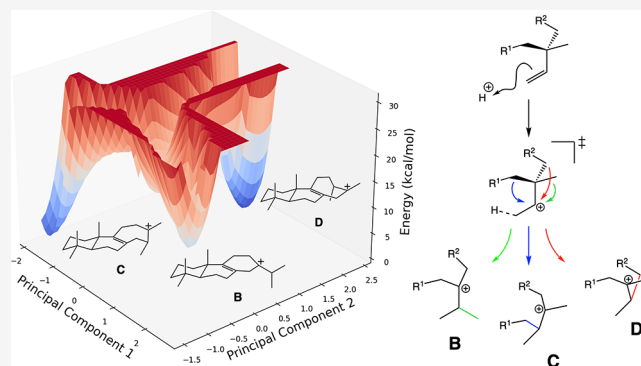


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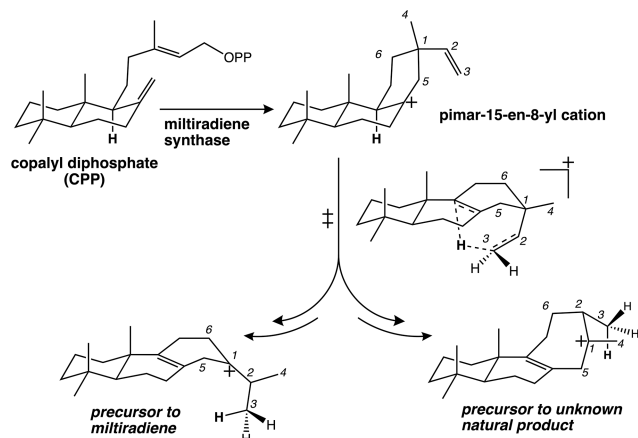
ABSTRACT: The role of post-transition state bifurcations in carbocation rearrangements derived from copalyl diphosphate (CPP) and its diastereomers *syn*-CPP and *ent*-CPP were investigated using quantum chemical calculations, *ab initio* molecular dynamics (AIMD) simulations, and potential energy surface (PES) analysis based on principal component analysis (PCA). Proton transfer transition states of pimarenyl cations, which can give rise to multiple products through bifurcating pathways, were the focus. For normal CPP-derived systems, AIMD simulations revealed that *Re*-face proton transfer leads to a diverse array of products, whereas *Si*-face proton transfer predominantly follows a single pathway. In contrast, *syn*-CPP and *ent*-CPP-derived systems showed reduced product diversity, with bifurcations only observed for the *Re*-face proton transfer. Orbital alignment analyses highlight the influence of dynamic momentum effects on product distributions. PES analyses further demonstrate that the CPP-derived system possesses the flattest energy landscape. These results highlight how simple stereochemical changes can dramatically alter dynamical outcomes, thereby modulating the challenges faced by enzymes in shaping product selectivity.



INTRODUCTION

Can a single transition state lead directly to more than one product? Yes, via a post-transition state bifurcation (PTSB).^{1–5}

Scheme 1. Potential PTSB in Miltiradiene Biosynthesis



Once thought to be curiosities of dubious “real world relevance”, PTSBs have been shown to be relevant to synthetic organic chemistry,^{1,2,6–10} homogeneous catalysis via organometallic species,^{11–13} and biosynthesis.^{1,8,14,15} In the lattermost context, PTSBs have been shown to be particularly relevant to

carbocation rearrangements proceeding along reaction coordinates with relatively flat regions (regions where structures change considerably but energy does not)—a scenario that has been encountered in a variety of terpene-forming reactions.^{15,16}

While most of the modeling of such reactions has focused on inherent carbocation reactivity,^{17,18} the inclusion of full terpene synthase enzymes has shown that this inherent reactivity is generally expressed (although sometimes modulated in degree).¹⁹ In particular, PTSBs have been shown to persist in the context of an explicitly modeled terpene synthase enzyme.^{20–22} Experiments also support the importance of inherent carbocation reactivity, leading to the view that those interested in terpene synthase mechanisms ought to consider the intrinsic tendencies of reactive carbocations to determine what role a surrounding enzyme actually plays in determining rate and/or selectivity.¹⁷

A particularly unusual reaction network containing multiple PTSBs was previously discovered as a potential biosynthetic

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Scheme 2. Diastereomers of CPP Examined

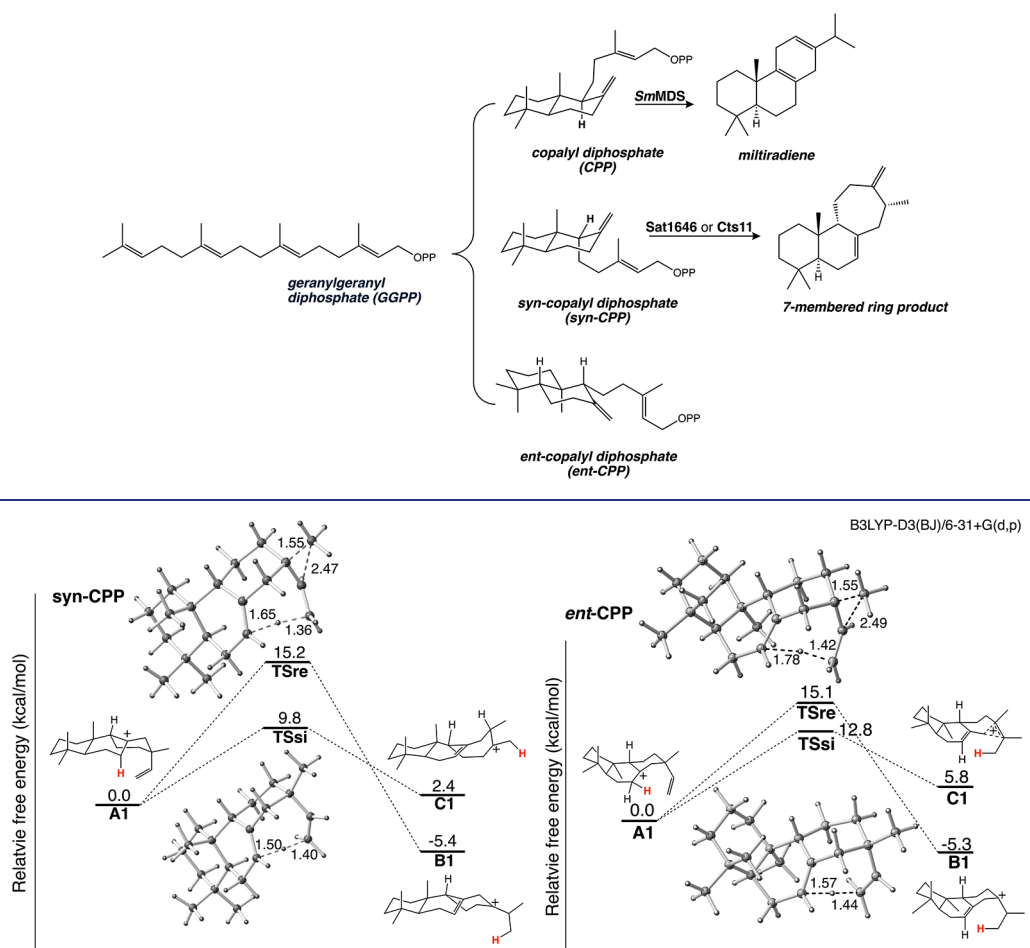


Figure 1. Computed energy profiles for both *Re* and *Si* proton transfers in *syn*-CPP and *ent*-CPP-derived pimarenyl cations.

route to miltiradiene (Scheme 1).¹⁴ In this case, a transition state for intramolecular proton transfer^{23,24} of the pimar-15-en-8-yl cation (which would be derived from copalyl diphosphate, CPP, in a miltiradiene synthase enzyme) was followed by a PTSB, which was followed by additional PTSBs and flat regions of the potential energy surface (PES), allowing for direct access to 6–12 product minima (the range is a result of differences resulting from using potential energy, free energy, or molecular dynamics trajectories). This discovery necessarily reshaped the questions one should ask about an enzyme making use of this type of transition state. Instead of asking how such an enzyme forms miltiradiene, one perhaps should ask how it avoids the formation of so many other easily accessible products.

One also might ask if any of these other products are formed in Nature. The inherent dynamical tendency of the pimar-15-en-8-yl cation in question was shown to form a carbocation precursor to miltiradiene preferentially,¹⁴ indicating that Nature might indeed take advantage of the dynamical properties of substrates (and opening a discussion on the role of these in enzyme evolution). The second most likely product, based on ab initio molecular dynamics (AIMD) simulations^{25–28} initiated from the proton transfer transition state, possessed a skeleton with a 7-membered ring (Scheme 1). While, to our knowledge, a natural product with this skeleton is still not known, a closely related skeleton has been described as

a minor product of terpene synthases that produce a stemodene and several pimaradienes (Scheme 2).^{29,30} This natural product would be derived from the *syn* diastereomer of CPP. The disclosure of this structure thus begs the following questions: Do pimar-15-en-8-yl cations derived from diastereomers of CPP also have transition states for intramolecular proton transfer followed by reaction pathway bifurcations? If so, what product selectivities are expected based on the inherent dynamical tendencies of the carbocations involved? If not, why not? What, if anything, is unique about the precursor to miltiradiene? It was shown previously that a diastereomeric proton transfer transition state derived from CPP (proton transfer to the *Si* rather than *Re* face of the alkene that accepts the proton) was not followed by a PTSB and did not have a tendency to form precursors to miltiradiene (which would have been possible, in principle, since proton transfer does not create a stereogenic center),¹⁴ suggesting that relatively small changes to structure could have dramatic consequences.

Here we investigated pimar-15-en-8-yl cations derived from diastereomers of CPP (Scheme 2) using quantum chemistry and AIMD simulations to determine whether or not PTSBs are present and why and why not. We analyzed the dynamic effects influencing predicted product distributions in terms of geometries and molecular orbitals, along with potential energy surfaces constructed using a new approach based on principal component analysis (PCA).

Scheme 3. Three Types of Major Products Expected to Result from Proton Transfer (left) and Structures for Non-target Products (right)

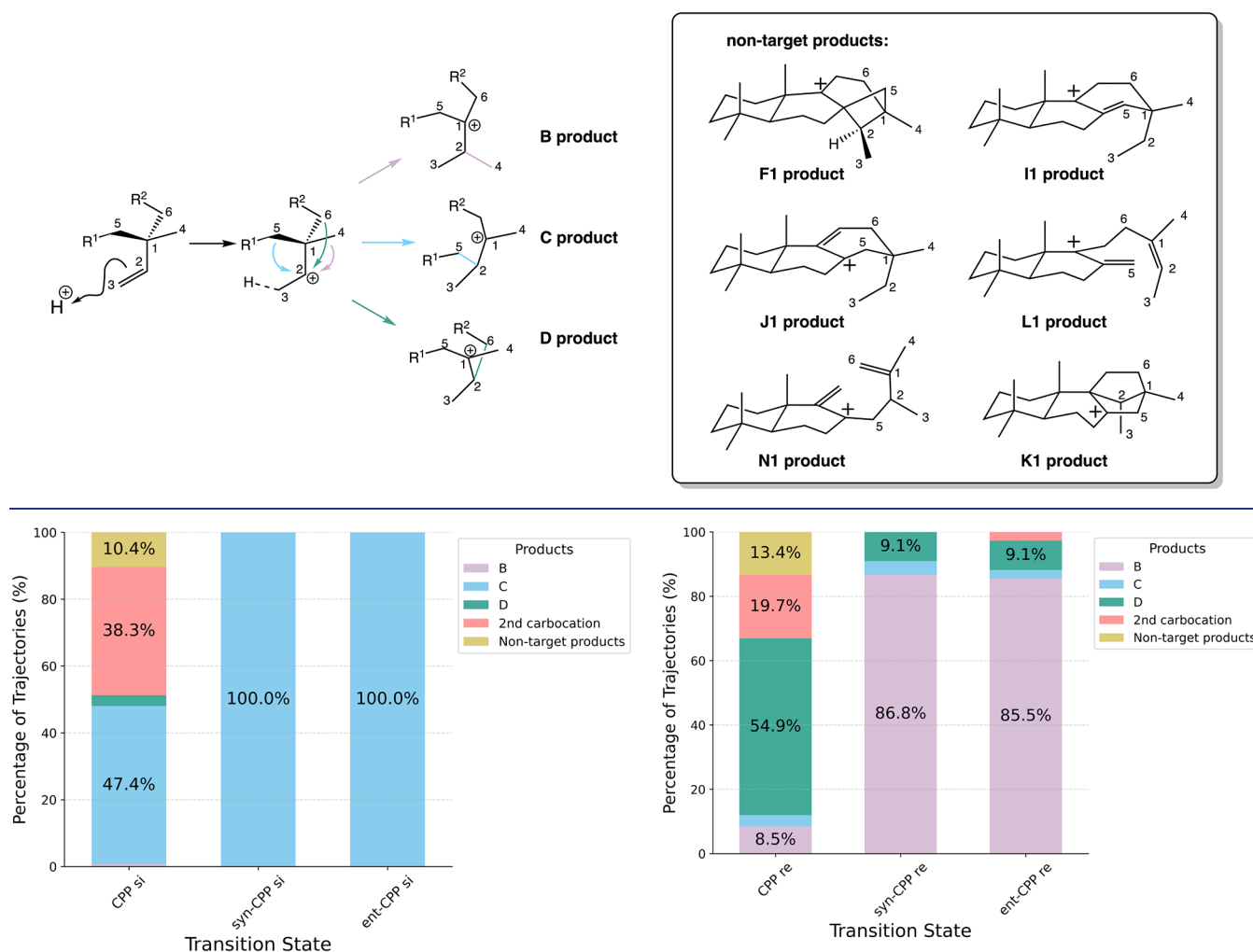


Figure 2. Distribution of trajectory outcomes (B3LYP-D3(BJ)/6-31G(d)) from different transition states for pimaranyl cations derived from CPP, *syn*-CPP, and *ent*-CPP. Results with *Si* transition states are at the left, while those from *Re* transition states are at the right.

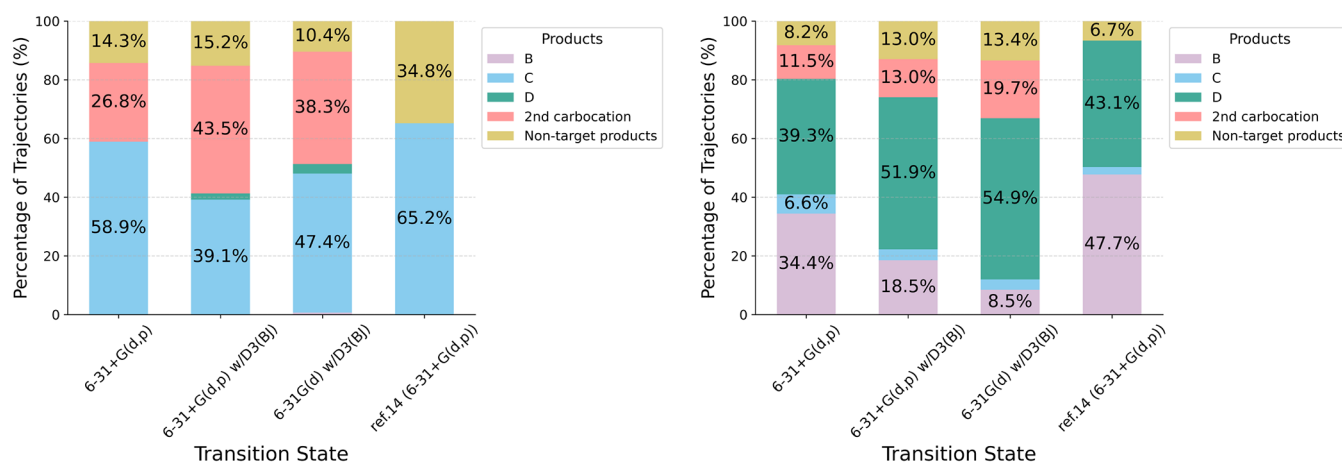


Figure 3. Distribution of trajectory outcomes from *Si* (left) and *Re* (right) transition states for pimaranyl cations derived from CPP using different levels of theory.

COMPUTATIONAL METHODS

All quantum chemical calculations were performed with *Gaussian16*.³¹ All geometries were optimized at the B3LYP-D3(BJ)/6-31+G(d,p) level of theory, which includes dispersion correction.^{32–38} Harmonic vibrational frequencies were calculated at the same level of theory. Intrinsic reaction coordinate (IRC) calculations were used to confirm the minima connected to the transition structures.^{39–41} All AIMD simulations were quasiclassical and were performed using Prody^{42,43} with trajectories initiated from transition states (using the B3LYP-D3(BJ)/6-31G(d) level of theory; we found that the reduced basis set gives qualitatively similar answers, while allowing for faster

MD simulations). PES energies were obtained from relaxed scans, and the surface coordinates were derived from PCA (scikit-learn; see the [Supporting Information](#) for additional details).⁴⁴

RESULTS AND DISCUSSION

Transition Structures and Barriers. While terpene synthase enzymes facilitate the disconnection of the substrate's diphosphate group, triggering a cyclization that generates a pimarenyl cation (e.g., [Scheme 1](#)), we focused here on the subsequent reaction pathways for the diastereomers of CPP shown in [Scheme 2](#). Two transition structures for intramolecular proton transfer were expected for each diastereomer since the vinyl group of each pimarenyl cation can abstract a proton with either its *Re* or *Si* face. The corresponding energy profiles for pimarenyl cations derived from *syn*- and *ent*-CPP are illustrated in [Figure 1](#). For both cases, two transition structures were found and *Si* proton transfer was preferred, as was the case for pimarenyl cations derived from CPP.¹⁴ The magnitude of the preference for *Si* proton transfer varied, however, which is presumably a result of different strain associated with differences in relative configuration. Note that for the diastereomers examined here, *Re* proton transfer is coupled to ring expansion and *Si* proton transfer is coupled to methyl shift based on IRC calculations. However, as described above, previous results indicated that transition states for proton transfer from CPP-derived cations lead to multiple products, an issue that MD simulations are required to address.

Dynamical Tendencies. AIMD simulations were performed for each system. Trajectories were initiated from both the *Si* and *Re* proton transfer transition states. The three main types of products that are formed for the pimarenyl cations examined here are denoted as **B** (6-membered ring), **C** (7-membered ring; migration from “front”), and **D** (7-membered ring; migration from “back”); all other products are considered to be “non-target products” ([Scheme 3](#)). Results are summarized in [Figure 2](#). See the [Supporting Information](#) for details on the distribution of trajectories forming each nontarget product. In the case of CPP product trajectories, many remain in the region of the secondary carbocation after 500 fs (labeled as the “2nd carbocation” in [Figures 2 and 3](#)) ([Scheme 3](#)).

Trajectories initiated from pimarenyl cations derived from normal CPP lead to more products than do their diastereomers—regardless of whether they originate from the *Si* or *Re* transition states. While the product distributions are not quantitatively the same as those found previously,¹⁴ the result of the different level of theory used here and variations expected in running modest numbers of trajectories ([Figure 3](#)), the significant amounts of products **B** and **D** were expected for the AIMD simulations initiated from the *Re* transition state, as was the formation of multiple products ([Figures 2 and 3](#), right). The product distribution found for the *Si* transition state is also similar to that found previously ([Figures 2 and 3](#), left).¹⁴ Note, however, that in the previously published work, trajectories were propagated past the secondary cation (sometimes for an additional ~1000 fs) to determine the final product, complicating comparisons.¹⁴

For both *syn*-CPP and *ent*-CPP-derived cations, only one product (the one connected to the transition structure by IRC) is formed from *Si* proton transfer ([Figure 2](#), left), indicating that there is no PTSB in this system. In contrast, *Re* proton transfers for these two diastereomers do exhibit bifurcations, but these have strong preferences for products **B** (again

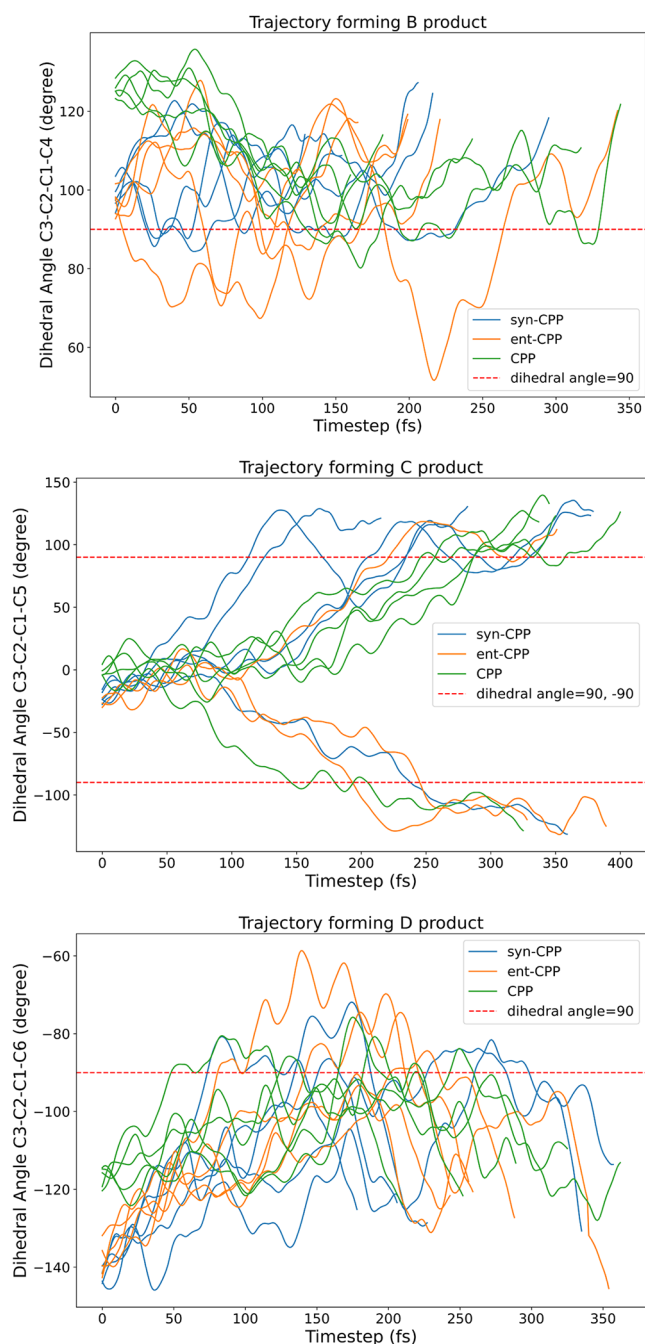


Figure 4. Evolution of key dihedral angles (C3–C2–C1–CX; X = 4 for B, X = 5 for C, and X = 6 for D) along trajectories leading to the formation of products B (top), C (middle), and D (bottom). Red dashed lines indicate the angle ($\sim 90^\circ$) ideal for orbital overlap of the migrating C1–CX bond and the formally empty p-orbital on C2.

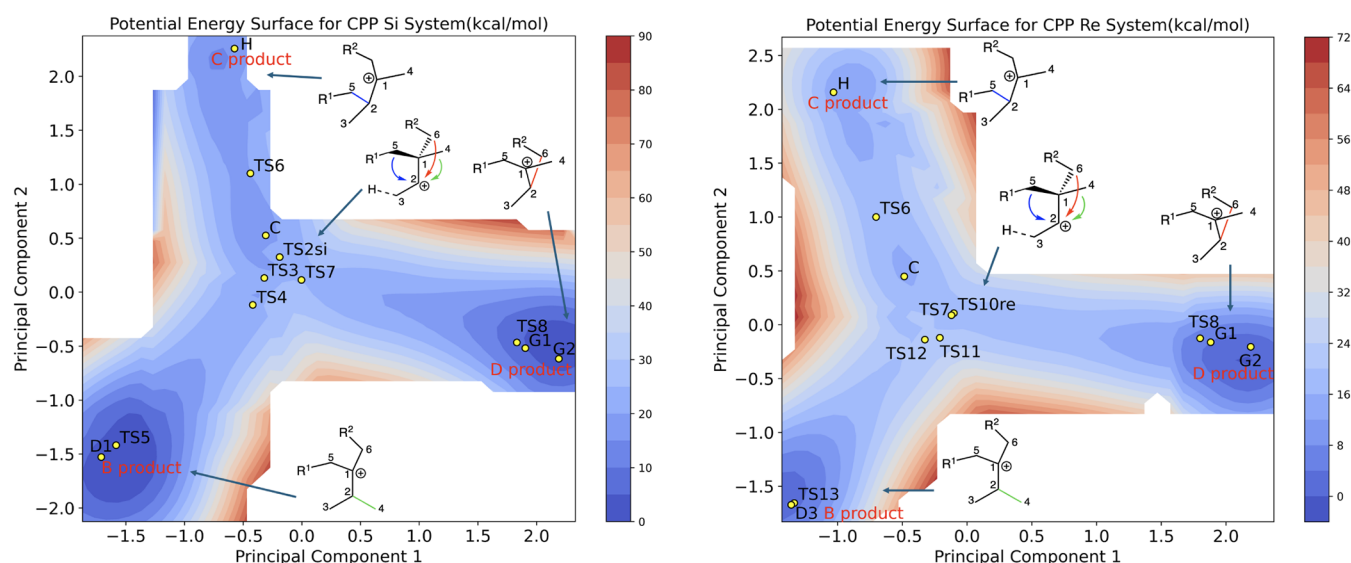


Figure 5. 2D contour plots of the PESs for *Si* (left) and *Re* (right) proton transfer in the CPP-derived systems. The color scale represents relative energy differences in kcal/mol. In both figures, structural labels are preserved from a previously published study to maintain consistency; for specific structural definitions, please refer to the [Supporting Information](#); labels in red denote the corresponding product class used in this study.

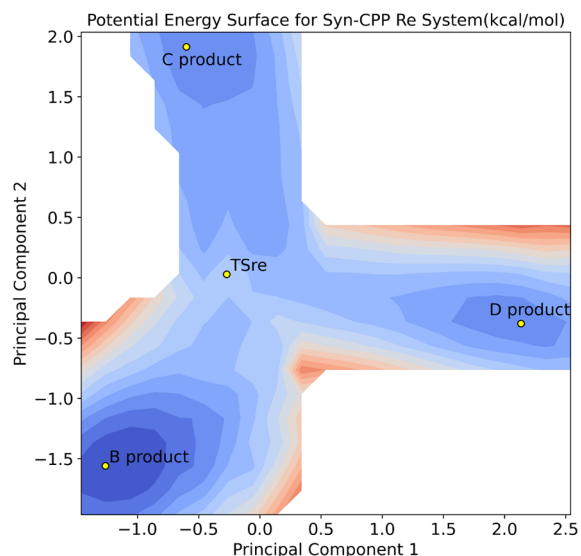


Figure 6. 2D contour plot of the PES for *Re* proton transfer in the *syn*-CPP-derived system.

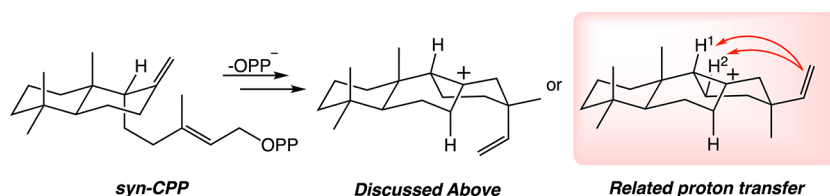
matching IRC predictions) (Figure 2, right). Thus, the CPP-based system does appear to be unusual in terms of its dynamic behavior, but why?

Trajectory Analysis. Since the migrating σ -bond must align with the formally empty p-orbital of the carbocation for migration to occur (see [Supporting Information](#) for NBO data), we tracked the C3–C2–C1–CX ($X = 4$ for **B**, $X = 5$ for

C, $X = 6$ for **D**) dihedral angles (Scheme 3) along trajectories (Figure 4) to examine whether appropriate orbital alignment was present when transition states were reached. Here, we focus our discussion on *Re* proton transfer since this leads to the most products. For each of CPP, *ent*-CPP, and *syn*-CPP, we randomly selected five trajectories leading to the formation of **B**, **C**, and **D** products. The evolution of the relevant dihedral angles along these trajectories is summarized in Figure 4. To achieve maximum orbital overlap for shifting, each dihedral angle must approach $\pm 90^\circ$. For trajectories leading to products **B**, dihedral angles for *ent*-CPP and *syn*-CPP have already approached 90° as the proton is transferred. However, for normal CPP, ~ 150 fs passes before this dihedral angle is reached. This difference is consistent with a momentum effect working against the formation of **B** products with CPP, the diastereomer for which such products are formed only in small amounts. For trajectories leading to products **C**, dihedral angles for all three systems are not near 90° as the proton is transferred, consistent with the low amounts of **C** products observed. For trajectories leading to products **D**, again dihedral angles for none of the three systems are close to 90° but those for the normal CPP system are closest, consistent with a momentum effect at least contributing to the increased amount of **D** products in this system.

Construction of Potential Energy Surfaces. To further investigate the dynamical tendencies of the reactive species in these reactions, we aimed to construct potential energy surfaces (PESs) and examine how their shapes influence dynamic selectivity. However, the numerous degrees of

Scheme 4. Proton Transfer Pathways of the CPP Carbocation



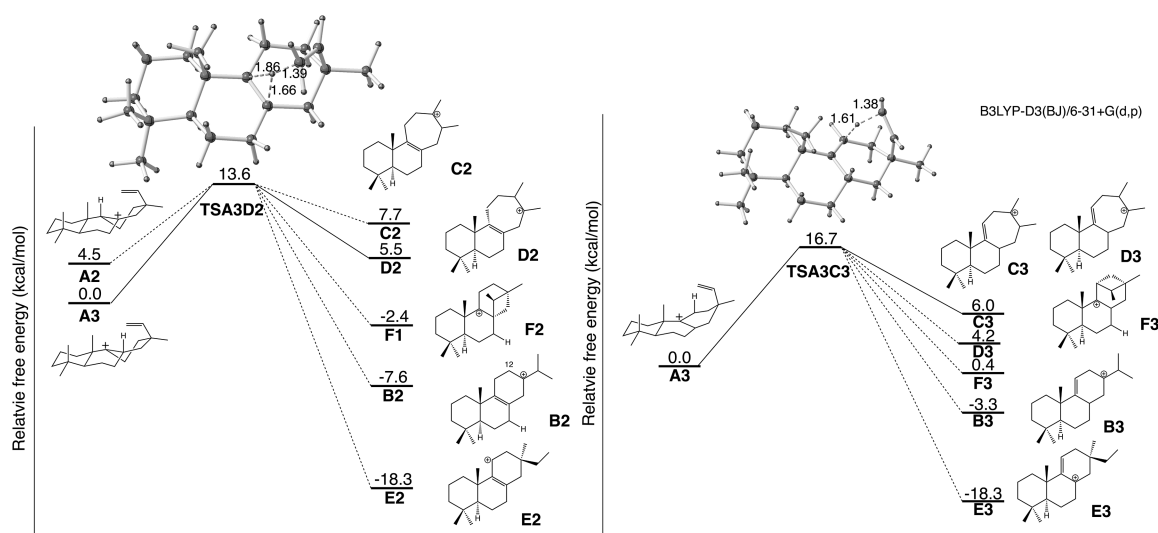


Figure 7. Optimized geometries of alternative transition states for proton transfer in the *syn*-CPP system. Solid lines indicate IRC-derived products, while dashed lines represent products obtained from the AIMD simulation.

Table 1. Trajectory Distributions from AIMD Simulations^a

	Reactants		Products					Recrossing
	A2	A3	B	C	D	E	F	
TSA3C3	0	59	22	33	1	1	2	5
TSA3D2	67	11	18	5	43	3	9	13

^aShown are the product outcomes for trajectories initiated from **TSA3C3** and **TSA3D2**.

freedom involved presented challenges in accurately mapping the PESs.^{45,46} To tackle this challenge, we employed PCA to reduce the dimensionality to two principal components.⁴⁷ Specifically, PCA was applied to all changing (breaking/forming) bond distances, allowing us to identify the two components that capture the most structural variation (see the [Supporting Information](#) for details). These components reflect linear combinations of the bond distance changes. The resulting two-dimensional coordinate space was then scanned to construct the PESs.

A comparison of *Si*- and *Re*-face proton transfer for the CPP-derived pimarenyl cation is shown in [Figure 5](#).¹⁴ Relaxed scans were performed at the B3LYP-D3(BJ)/6-31G(d) level. While the two principal components do not fully capture all structural differences—some related structures such as **TS8** and **G1** are difficult to distinguish on this PES—the PES still captures features useful for interpreting the results of molecular dynamics simulations. As shown in [Figure 2](#), trajectories initiated from the transition state for *Si*-face proton transfer predominantly yielded products **C**, whereas those initiated from the transition state for *Re*-face proton transfer more often produced products **B** and **D**. The formation of products **C** from the *Si* transition structure is consistent with this transition structure (labeled **TS2** here based on the previous study¹⁴) being close to the shallow secondary cation intermediate (coincidentally called “C” in the previous study¹⁴ and the exit channel to products **C**. A substantial number of AIMD trajectories initiated from the *Si* transition state also do not escape from the secondary cation region after 500 fs. The preferential formation of products **B** and **D** from the *Re*

transition state is not as straightforward to rationalize, but the flatness of the PES in the vicinity of the transition structure is consistent with momentum playing a large role, as previously postulated.¹⁴ In contrast, the PES for *Re* proton transfer from the *syn*-CPP-derived pimarenyl cation ([Figure 6](#)) is far less flat.

Related Proton Transfers. To further explore how structural differences influence the intrinsic reactivity of different CPP diastereomers, we extended our analysis to include possible additional proton transfer pathways in the *syn*-CPP system ([Scheme 4](#)). For both reactions, transition structures leading to multiple products are found. For H¹ deprotonation, AIMD trajectories indicate bifurcating pathways in both the reactant and product directions. [Figure 7](#) (left) shows the transition structures and all reactants and minima connected to them by either IRC or AIMD calculations. Corresponding product distributions from AIMD simulations are summarized in [Table 1](#). While **TSA3C3** leads predominantly to two products, **TSA3D2** leads to multiple products, whose relative amounts do not reflect their relative energies. This latter scenario corresponds to comparable diversity as observed previously for *Re* proton transfer with normal CPP, but with lower inherent dynamical selectivity!

Implications. The combination of calculations on energy surfaces and molecular dynamics trajectories described here for intramolecular proton transfers in diastereomeric pimar-15-en-8-yl carbocations has revealed that post-transition state bifurcations are common for such molecular architectures, but the associated product diversity and inherent dynamical selectivity vary greatly. Consequently, the challenges faced by terpene synthase enzymes chaperoning such chemistry will differ from system to system, with some systems requiring more intervention than others if high selectivity is functionally useful. Of course, product diversity also has utility,⁴⁸ and here we show how some systems are perhaps better suited to that than others. We believe strongly that attempts to predict terpene synthase function (from sequence or otherwise) should take inherent substrate reactivity into account,¹⁷ and that reactivity often involves nonstatistical dynamic effects given the nature of delocalization and flat PESs associated with

carbocations.¹⁵ Enzymes do remarkable things, but so do their substrates!

ASSOCIATED CONTENT

* Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.5c16476>.

Additional computational details, including methods, benchmarking results, NBO analysis, IRC plots (PDF)

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Notes

The authors declare no competing financial interest.

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